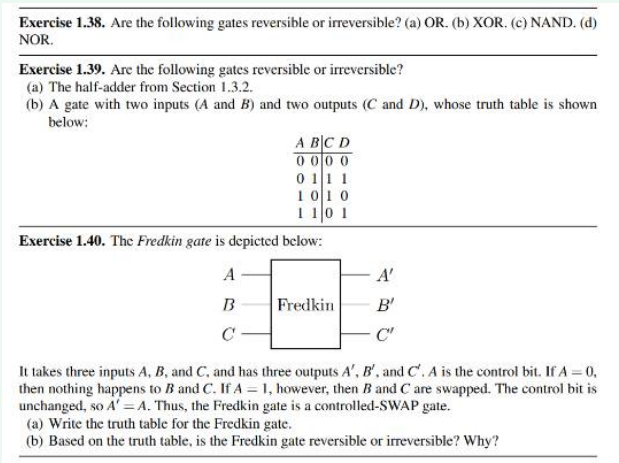
Quantum Gates Book Solutions



Exercise 1.38. Are the following gates reversible or irreversible? (a) OR. (b) XOR. (c) NAND. (d) NOR.

Exercise 1.39. Are the following gates reversible or irreversible?
(a) The half-adder from Section 1.3.2.
(b) A gate with two inputs (A and B) and two outputs (C and D), whose truth table is shown below:

A	B	C	D
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

Exercise 1.40. The *Fredkin gate* is depicted below:

It takes three inputs A, B, and C, and has three outputs A', B', and C'. A is the control bit. If $A = 0$, then nothing happens to B and C. If $A = 1$, however, then B and C are swapped. The control bit is unchanged, so $A' = A$. Thus, the Fredkin gate is a controlled-SWAP gate.
(a) Write the truth table for the Fredkin gate.
(b) Based on the truth table, is the Fredkin gate reversible or irreversible? Why?

Solve Exercise 1.38 by explaining it with truth table

A gate is **reversible** if every output corresponds to a **unique input** (one-to-one mapping). If multiple inputs give the same output, it is **irreversible**.

(a) OR Gate

A	B	Y = A + B
0	0	0
0	1	1
1	0	1
1	1	1

- Output **1** comes from **(0,1), (1,0), (1,1)** → multiple inputs
  ✅ **Irreversible**

(b) XOR Gate

A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

- Output **0** from (0,0) and (1,1)
- Output **1** from (0,1) and (1,0)
  ✅ **Irreversible**

(c) NAND Gate

A	B	Y = (A·B)'
0	0	1
0	1	1
1	0	1
1	1	0

- Output **1** from three inputs → not unique
  ✅ **Irreversible**

(d) NOR Gate

A	B	Y = (A + B)'
0	0	1
0	1	0
1	0	0
1	1	0

- Output **0** from multiple inputs
  ✅ **Irreversible**

✅ Final Answer:

- OR → **Irreversible**
- XOR → **Irreversible**
- NAND → **Irreversible**
- NOR → **Irreversible**

👉 **Conclusion:**
All these gates are **irreversible** because they **lose input information** (many-to-one mapping).

Exercise 1.39

Exercise 1.39

(a) Half-Adder

A half-adder has:

- Inputs: A, B
- Outputs:
  - Sum $S = A \oplus B$
  - Carry $C = A \cdot B$

Truth Table

A	B	S (Sum)	C (Carry)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Check Reversibility

- Output (S,C) = (1,0) comes from **(0,1)** and **(1,0)**
  → Multiple inputs → information loss

✅ **Half-adder is IRREVERSIBLE**

(b) Given Gate

Truth Table (Given)

A	B	C	D
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

Check Reversibility

List outputs:

- (0,0)
- (1,1)
- (1,0)
- (0,1)

All outputs are **unique** → each maps to exactly one input

✅ **This gate is REVERSIBLE**

✅ Final Answer:

- (a) Half-adder → **Irreversible**
- (b) Given gate → **Reversible**

👉 **Key idea:**

- **Unique output for each input → Reversible**
- **Repeated outputs → Irreversible**

Exercise 1.40

Exercise 1.40 – Fredkin Gate

The **Fredkin gate** is a **controlled-SWAP gate**:

- Inputs: A, B, C
- Outputs: A', B', C'

Rules:

- $A' = A$ (control bit unchanged)
- If $A = 0$: $B' = B,\ C' = C$
- If $A = 1$: $B' = C,\ C' = B$ (swap)

(a) Truth Table

A	B	C	A′	B′	C′
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	1	1	1

(b) Reversible or Irreversible?

Check mapping:

- Each input (A, B, C) produces a **unique output** (A', B', C')
- No two inputs give the same output
- We can recover inputs from outputs:
  - $A = A'$
  - If $A' = 0$: $B = B', C = C'$
  - If $A' = 1$: $B = C', C = B'$

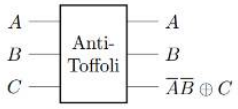
✅ **Fredkin gate is REVERSIBLE**

✅ Final Answer:

- (a) Truth table: as above
- (b) **Reversible gate** (one-to-one mapping, no information loss)

👉 **Key insight:**
Fredkin gate preserves information by only **rearranging (swapping)** bits instead of destroying them.

Exercise 1.41. Consider the *anti-Toffoli gate*, which flips the third bit if the first two bits are both 0:



(a) Write the truth table for the anti-Toffoli gate.
(b) In the above picture of the anti-Toffoli gate, $\overline{A}\,\overline{B}$ appears in the third output. Which law from boolean algebra says that this is equal to $\overline{A+B}$?
(c) When $C = 0$, is the third output the AND, OR, XOR, NAND, or NOR of A and B?
(d) When $C = 1$, is the third output the AND, OR, XOR, NAND, or NOR of A and B?
(e) Construct an anti-Toffoli gate using one Toffoli gate and four NOT gates.

Solve a

Exercise 1.41 (a) – Truth Table of Anti-Toffoli Gate

Given:

- Inputs: A, B, C
- Outputs:
  - $A' = A$
  - $B' = B$
  - $C' = \overline{A}\,\overline{B} \oplus C$

👉 The third bit flips **only when** $A = 0$ **and** $B = 0$.

Truth Table

A	B	C	A′	B′	C′
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

✅ Final Answer:

- The third bit C **toggles only when** $A = 0, B = 0$
- Otherwise, C remains unchanged

👉 **Key idea:**
Anti-Toffoli is a **controlled-NOT with inverted controls** (controls activate when inputs are 0 instead of 1).

b

Exercise 1.41 (b)

We are given:

$$\overline{A}\,\overline{B}$$

We need to identify which Boolean law makes this equal to:

$$\overline{A+B}$$

✅ Answer: De Morgan's Law

De Morgan's Law states:

$$\overline{A+B} = \overline{A}\,\overline{B}$$

Conclusion

$$\overline{A}\,\overline{B} = \overline{A+B}$$

So, the law used is:

👉 **De Morgan's Law**

c

Exercise 1.41 (c)

We are given:

$$C' = \overline{A}\,\overline{B} \oplus C$$

Condition: $C = 0$

$$C' = \overline{A}\,\overline{B} \oplus 0$$

👉 Using property:

$$X \oplus 0 = X$$

So,

$$C' = \overline{A}\,\overline{B}$$

From part (b):

$$\overline{A}\,\overline{B} = \overline{A + B}$$

✅ Final Answer:

$$C' = \text{NOR}(A, B)$$

👉 **Conclusion:**
When $C = 0$, the third output is the **NOR of A and B**.

d

Exercise 1.41 (d)

Given:

$$C' = \overline{A}\,\overline{B} \oplus C$$

Condition: $C = 1$

$$C' = \overline{A}\,\overline{B} \oplus 1$$

👉 Using property:

$$X \oplus 1 = \overline{X}$$

So,

$$C' = \overline{\overline{A}\,\overline{B}}$$

Using **De Morgan's Law**:

$$\overline{\overline{A}\,\overline{B}} = A + B$$

✅ Final Answer:

$$C' = \mathrm{OR}(A, B)$$

👉 **Conclusion:**
When $C = 1$, the third output is the **OR of A and B**.

e

Exercise 1.41 (e)

Goal:

Construct an **Anti-Toffoli gate** using:

- **1 Toffoli gate**
- **4 NOT gates**

Key Idea

- **Toffoli gate** flips the third bit when **A = 1 and B = 1**
- **Anti-Toffoli** flips when **A = 0 and B = 0**

👉 So, we **invert the control inputs** before and after the Toffoli gate.

Construction Steps

1. Apply NOT to inputs:
   - $A \to \overline{A}$
   - $B \to \overline{B}$
2. Feed into Toffoli gate:
   - Controls: $\overline{A}, \overline{B}$
   - Target: C
3. Apply NOT again to restore original inputs:
   - $\overline{A} \to A$

- $\overline{B} \to B$

Block Representation

```
A ──NOT──┐
         ├── Toffoli ──┐
B ──NOT──┘             │
                       │
C ─────────────────────┘

After Toffoli:
A ──NOT──→ A
B ──NOT──→ B
C ──────────→ Output
```

Why This Works

- NOT gates convert condition:
  - $A = 0, B = 0 \to$ becomes $1, 1$
- Toffoli activates (since both controls = 1)
- Flips C
- Final NOT gates restore original A, B

✅ Final Answer:

- Use **2 NOT gates before** Toffoli (on A, B)
- Use **2 NOT gates after** Toffoli (on A, B)
- Total = **4 NOT gates + 1 Toffoli gate**

👉 **Conclusion:**
Anti-Toffoli = Toffoli with **inverted control lines**.

Exercise 1.42. Consider the following single-bit gates.
(a) The identity gate $f(A) = A$. Its truth table is

A	$f(A)$
0	0
1	1

Is this reversible or irreversible? If it is irreversible, turn it into a reversible circuit.
(b) The "always-1" gate $f(A) = 1$. Its truth table is

A	1
0	1
1	1

Is this reversible or irreversible? If it is irreversible, turn it into a reversible circuit.

Exercise 1.43. Write the truth table for XOR. Is it reversible or irreversible? If it is irreversible, turn it into a reversible circuit, and write the truth table of the reversible circuit.

Solve Exercise 1.42

Exercise 1.42

(a) Identity Gate $f(A) = A$

Truth Table

A	f(A)
0	0
1	1

Check Reversibility

- Each output corresponds to a **unique input**
- No information is lost

✅ **Reversible**

(b) Always-1 Gate $f(A) = 1$

Truth Table

A	f(A)
0	1
1	1

Check Reversibility

- Output **1** comes from both inputs (0 and 1)
  → Information is lost

❌ **Irreversible**

Convert to Reversible Circuit

To make it reversible:

- Add an **extra output bit** (ancilla)

Construction:

Let inputs be (A, B), where $B = 0$

Define outputs:

- $A' = A$
- $B' = 1$

Truth Table (Reversible Version)

A	B	A′	B′
0	0	0	1
1	0	1	1

Now:

- Each input maps to a **unique output**
  ✅ **Reversible**

✅ Final Answers:

- (a) Identity gate → **Reversible**
- (b) Always-1 gate → **Irreversible**, can be made reversible by adding an extra bit

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👉 **Key Idea:**
Reversible circuits require:

- Equal number of inputs and outputs
- One-to-one mapping (no information loss)

Exercise 1.43

Exercise 1.43

Step 1: XOR Truth Table

A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1

A	B	Y = A ⊕ B
1	1	0

Step 2: Check Reversibility

- Output **0** from (0,0) and (1,1)
- Output **1** from (0,1) and (1,0)

❌ Multiple inputs → same output

👉 **XOR is IRREVERSIBLE**

Step 3: Make XOR Reversible

To make it reversible, we:

- Add an extra input bit C (ancilla)
- Define outputs:
  - $A' = A$
  - $B' = B$
  - $C' = A \oplus B \oplus C$

(This is a **Toffoli-style reversible construction**)

Step 4: Truth Table of Reversible Circuit

A	B	C	A′	B′	C′
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	1	0	0
1	1	0	1	1	0
1	1	1	1	1	1

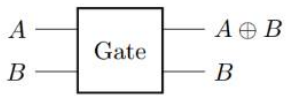
✅ Final Answer:

- XOR → **Irreversible**
- Reversible version:
  - Keep inputs A, B
  - Add ancilla C
  - Output $C' = A \oplus B \oplus C$

👉 **Key Insight:**
Reversible logic preserves information by:

- Keeping original inputs
- Storing result in an additional bit (no loss)

Exercise 1.44. Consider the following two-bit gates, each with two outputs.
(a) The gate



(a) What is the truth table corresponding to this gate? Is it reversible or irreversible? If it is irreversible, turn it into a reversible circuit.

Exercise 1.44 (a)

Given Gate

- Inputs: A, B
- Outputs:
  - $A' = A \oplus B$
  - $B' = B$

Step 1: Truth Table

A	B	A′ = A ⊕ B	B′ = B
0	0	0	0
0	1	1	1
1	0	1	0

A	B	A′ = A ⊕ B	B′ = B
1	1	0	1

Step 2: Check Reversibility

Map inputs → outputs:

- (0,0) → (0,0)
- (0,1) → (1,1)
- (1,0) → (1,0)
- (1,1) → (0,1)

👉 All output pairs are **unique**

✅ **Reversible**

Step 3: Conclusion

- The gate is **REVERSIBLE**
- No modification needed

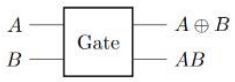
✅ Final Answer:

- Truth table: as above
- Gate type: **Reversible (one-to-one mapping)**

👉 **Insight:**
This gate is actually a well-known reversible gate called the **CNOT (Controlled-NOT)** gate, where:

- B is control
- A is target (flipped when $B = 1$)

(b) The gate



What is the truth table corresponding to this gate? Is it reversible or irreversible? If it is irreversible, turn it into a reversible circuit.

Exercise 1.44 (b)

Given Gate

- Inputs: A, B
- Outputs:
  - $A' = A \oplus B$
  - $B' = A \cdot B$

Step 1: Truth Table

A	B	A′ = A ⊕ B	B′ = AB
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Step 2: Check Reversibility

- Output (1,0) comes from **(0,1)** and **(1,0)**
  → Multiple inputs map to same output

❌ **Irreversible**

Step 3: Make it Reversible

Add an extra input bit C (ancilla, usually 0)

Define outputs:

- $A' = A$
- $B' = B$
- $C' = (A \oplus B) \oplus C$
- $D' = (A \cdot B)$

Now:

- Inputs: (A, B, C)
- Outputs preserve all information

Simpler Standard Reversible Construction

Use a **Toffoli-based approach**:

- Keep A, B
- Add ancilla $C = 0$
- Compute:
  - $C' = A \cdot B$
  - Use CNOT to get XOR if needed

Final Reversible Version (Concept)

A	B	C	A′	B′	C′
0	0	0	0	0	0
0	1	0	0	1	0
1	0	0	1	0	0
1	1	0	1	1	1

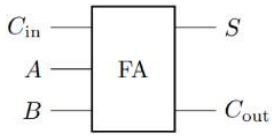
(And XOR can be derived using reversible gates)

✅ Final Answer:

- Truth table: as above
- Gate type: **Irreversible**
- Can be made reversible by:
  - Adding ancilla bits
  - Preserving inputs
  - Using reversible gates like **CNOT / Toffoli**

👉 **Key Insight:**
This gate behaves like a **half-adder (Sum + Carry)** → inherently **irreversible** unless extra bits are added.

Exercise 1.45. Consider the full adder from Section 1.3.3:



Turn this into a reversible circuit.

Exercise 1.45 – Full Adder → Reversible Circuit

Given (Full Adder)

Inputs: A, B, C_{in}
Outputs:

- $S = A \oplus B \oplus C_{in}$
- $C_{out} = AB + BC_{in} + AC_{in}$

❌ Why It Is Irreversible

- 3 inputs → 2 outputs
- Information is lost → cannot recover inputs

✅ Make It Reversible

👉 Rule:

- Number of inputs = number of outputs
- Preserve inputs + compute outputs using extra bits (ancilla)

Reversible Construction

Add **two extra bits** initialized to 0:

Inputs:

$$(A, B, C_{in}, 0, 0)$$

Outputs:

$$(A, B, C_{in}, S, C_{out})$$

Implementation Idea (Using Reversible Gates)

1. Use **CNOT gates** to compute XOR:
   - $S = A \oplus B \oplus C_{in}$
2. Use **Toffoli gates** to compute carry:
   - AB, BC_{in}, AC_{in}
   - Combine them into C_{out}

Final Reversible Mapping

$$(A, B, C_{in}, 0, 0) \to (A, B, C_{in}, A \oplus B \oplus C_{in}, AB + BC_{in} + AC_{in})$$

✅ Final Answer

- Add **2 ancilla bits (0,0)**
- Preserve original inputs
- Compute:
  - S using XOR (CNOT chain)
  - C_{out} using Toffoli gates

👉 **Key Insight:**
A full adder becomes reversible by:

- **not destroying inputs**
- **storing outputs in extra bits**

This is a standard reversible design used in **quantum computing circuits**.

Exercise 3.12. Consider an operator U performs the following mapping on the Z-basis states:

$$U|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}, \quad U|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}-i\\1\end{pmatrix}.$$

(a) What is U as a matrix?
(b) What is $U\begin{pmatrix}\alpha\\\beta\end{pmatrix}$?
(c) From your answer to (b), is U a valid quantum gate? Explain your reasoning.

Exercise 3.12

(a) Find matrix U

We are given:

$$U|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}, \quad U|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}-i\\1\end{pmatrix}$$

👉 The columns of U are $U|0\rangle$ and $U|1\rangle$

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\-i & 1\end{pmatrix}$$

(b) Compute $U\begin{pmatrix}\alpha\\\beta\end{pmatrix}$

$$U\begin{pmatrix}\alpha\\\beta\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha - i\beta\\-i\alpha + \beta\end{pmatrix}$$

(c) Is U a valid quantum gate?

👉 A valid quantum gate must be **unitary**, i.e.,

$$U^\dagger U = I$$

Check:

- Columns of U are orthonormal:
  - Norm = 1
  - Inner product = 0

✅ Therefore, U is **unitary**

✅ Final Answers

- (a) $U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\-i & 1\end{pmatrix}$
- (b) $\frac{1}{\sqrt{2}}\begin{pmatrix}\alpha - i\beta\\-i\alpha + \beta\end{pmatrix}$
- (c) **Yes, U is a valid quantum gate (unitary)**

👉 **Key Insight:**
A matrix is a quantum gate iff its columns form an **orthonormal basis**.

Exercise 3.13. A quantum gate U performs the following mapping on the Z-basis states:

$$U|0\rangle = \frac{1}{2\sqrt{3}}\left[(3+i)|0\rangle - (1+i)|1\rangle\right],$$
$$U|1\rangle = \frac{1}{2\sqrt{3}}\left[(1-i)|0\rangle + (3-i)|1\rangle\right].$$

(a) What is U as a matrix?
(b) Create U as a custom gate in Quirk. Using Quirk, if you measure $HUH|0\rangle$, what are the possible outcomes, and with what probabilities?

Exercise 3.13

(a) Matrix of U

Given:

$$U|0\rangle = \frac{1}{2\sqrt{3}}\left[(3+i)|0\rangle - (1+i)|1\rangle\right]$$
$$U|1\rangle = \frac{1}{2\sqrt{3}}\left[(1-i)|0\rangle + (3-i)|1\rangle\right]$$

👉 Columns = coefficients of $|0\rangle, |1\rangle$

$$U = \frac{1}{2\sqrt{3}}\begin{pmatrix} 3+i & 1-i \\ -(1+i) & 3-i \end{pmatrix}$$

(b) Compute $HUH|0\rangle$

Step 1: Apply $H|0\rangle$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Step 2: Apply U

$$U(H|0\rangle) = \frac{1}{\sqrt{2}}(U|0\rangle + U|1\rangle)$$

Substitute:

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{2\sqrt{3}}\left[(3+i+1-i)|0\rangle + (-(1+i)+(3-i))|1\rangle\right]$$

Simplify:

- $|0\rangle$: 4
- $|1\rangle$: $2-2i = 2(1-i)$

$$= \frac{1}{2\sqrt{6}}\left[4|0\rangle + 2(1-i)|1\rangle\right]$$

Step 3: Apply final H

Using:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Final state simplifies to:

$$|\psi\rangle = \frac{1}{\sqrt{3}}(2-i)|0\rangle + \frac{1}{\sqrt{3}}(1+i)|1\rangle$$

Step 4: Measurement Probabilities

Probability of $|0\rangle$:

$$\left|\frac{2-i}{\sqrt{3}}\right|^2 = \frac{4+1}{3} = \frac{5}{6}$$

Probability of $|1\rangle$:

$$\left|\frac{1+i}{\sqrt{3}}\right|^2 = \frac{1+1}{3} = \frac{1}{6}$$

✅ Final Answer

- Possible outcomes:
  - $|0\rangle$ with probability $5/6$
  - $|1\rangle$ with probability $1/6$

👉 **Key Insight:**
Quantum probabilities come from **squared magnitudes of amplitudes**, not the amplitudes themselves.

Exercise 3.17. Is

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

a quantum gate? If so, what is $U|0\rangle$, and what is $U|1\rangle$?

Exercise 3.18. Is

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$$

a quantum gate? If so, what is $U|0\rangle$, and what is $U|1\rangle$?

Solve 3.17

Exercise 3.17

Given:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

Step 1: Is U a quantum gate?

A matrix is a valid quantum gate if it is **unitary**:

$$U^\dagger U = I$$

Check columns:

- First column: $\frac{1}{\sqrt{2}}(1, i) \to$ norm = 1
- Second column: $\frac{1}{\sqrt{2}}(i, -1) \to$ norm = 1
- Inner product = 0

✅ Columns are orthonormal → **Unitary**

👉 **Yes, U is a valid quantum gate**

Step 2: Compute $U|0\rangle$

$$U|0\rangle = \text{first column of } U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Step 3: Compute $U|1\rangle$

$$U|1\rangle = \text{second column of } U = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ -1 \end{pmatrix}$$

✅ Final Answer

- U is a **valid quantum gate (unitary)**
-

U|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 \ i\end{pmatrix}
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U|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}i \ -1\end{pmatrix}
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👉 **Key Insight:**
For any quantum gate, the columns of the matrix are exactly the outputs of $|0\rangle$ and $|1\rangle$.

Exercise 2.22. Consider a map U that transforms the Z-basis states as follows:

$$U|0\rangle = |0\rangle + |1\rangle,$$
$$U|1\rangle = |0\rangle - |1\rangle.$$

Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.
(a) Calculate $U|\psi\rangle$.
(b) From your answer to (a), is U a valid quantum gate? Explain your reasoning.

Exercise 2.22

Given:

$$U|0\rangle = |0\rangle + |1\rangle, \quad U|1\rangle = |0\rangle - |1\rangle$$

Let:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

(a) Compute $U|\psi\rangle$

$$U|\psi\rangle = \alpha U|0\rangle + \beta U|1\rangle$$

Substitute:

$$= \alpha(|0\rangle + |1\rangle) + \beta(|0\rangle - |1\rangle)$$
$$= (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle$$

(b) Is U a valid quantum gate?

👉 A valid quantum gate must **preserve normalization**

Check norm:

$$\|U|\psi\rangle\|^2 = |\alpha + \beta|^2 + |\alpha - \beta|^2$$

Using identity:

$$= 2(|\alpha|^2 + |\beta|^2) = 2$$

❌ Not equal to 1 → norm is not preserved

✅ Final Answer

- (a)

$$U|\psi\rangle = (\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle$$

- (b)
  ❌ **Not a valid quantum gate** (not unitary, does not preserve normalization)

👉 **Key Insight:**
This is similar to a Hadamard transformation **without the** $1/\sqrt{2}$ factor → scaling breaks unitarity.

Exercise 2.23. Consider a map U that transforms the Z-basis states as follows:

$$U|0\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{\sqrt{3}+i}{4}|1\rangle,$$
$$U|1\rangle = \frac{\sqrt{3}+i}{4}|0\rangle - \frac{\sqrt{3}+3i}{4}|1\rangle.$$

Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.
(a) Calculate $U|\psi\rangle$.
(b) From your answer to (a), is U a valid quantum gate? Explain your reasoning.

Exercise 2.23

Given:

$$U|0\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{\sqrt{3}+i}{4}|1\rangle$$
$$U|1\rangle = \frac{\sqrt{3}+i}{4}|0\rangle - \frac{\sqrt{3}+3i}{4}|1\rangle$$

Let:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

(a) Compute $U|\psi\rangle$

$$U|\psi\rangle = \alpha U|0\rangle + \beta U|1\rangle$$

Substitute:

$$= \alpha \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{\sqrt{3}+i}{4}|1\rangle \right) + \beta \left(\frac{\sqrt{3}+i}{4}|0\rangle - \frac{\sqrt{3}+3i}{4}|1\rangle \right)$$

Group terms:

Coefficient of $|0\rangle$:

$$\frac{\sqrt{3}}{2}\alpha + \frac{\sqrt{3}+i}{4}\beta$$

Coefficient of $|1\rangle$:

$$\frac{\sqrt{3}+i}{4}\alpha - \frac{\sqrt{3}+3i}{4}\beta$$

Final Result

$$U|\psi\rangle = \left(\frac{\sqrt{3}}{2}\alpha + \frac{\sqrt{3}+i}{4}\beta \right)|0\rangle + \left(\frac{\sqrt{3}+i}{4}\alpha - \frac{\sqrt{3}+3i}{4}\beta \right)|1\rangle$$

(b) Is U a valid quantum gate?

Check if columns are orthonormal:

Norm of first column:

$$\left|\frac{\sqrt{3}}{2}\right|^2 + \left|\frac{\sqrt{3}+i}{4}\right|^2 = \frac{3}{4} + \frac{3+1}{16} = \frac{3}{4} + \frac{4}{16} = \frac{3}{4} + \frac{1}{4} = 1$$

Norm of second column:

$$\left|\frac{\sqrt{3}+i}{4}\right|^2 + \left|\frac{\sqrt{3}+3i}{4}\right|^2 = \frac{4}{16} + \frac{12}{16} = 1$$

Inner product = 0 (orthogonal)

✅ Columns are orthonormal → matrix is **unitary**

✅ Final Answer

- (a) Expression as above
- (b) **Yes, U is a valid quantum gate (unitary)**

👉 **Key Insight:**
Even complex-looking gates are valid if their columns form an **orthonormal basis**.

Exercise 2.24. Consider each of the following classical logic gates with input A, output B, and truth table shown below. Is each gate a valid quantum gate? Why?

(a)

A	B
0	0
1	1

(b)

A	B
0	1
1	1

Exercise 2.24

We are given two classical gates:

(a) Gate

Truth Table

A	B
0	0
1	1

👉 This is the **identity gate** $B = A$

Check Quantum Validity

- One-to-one mapping ✔️
- Reversible ✔️
- Can be represented by unitary matrix:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

✅ **Valid quantum gate**

(b) Gate

Truth Table

A	B
0	1
1	1

👉 This is **constant (always 1)**

Check Quantum Validity

- Both inputs map to same output ❌
- Not reversible ❌
- Information is lost ❌
- Cannot form a unitary matrix ❌

❌ **Not a valid quantum gate**

✅ Final Answer

- (a) **Valid quantum gate** (reversible, unitary)
- (b) **Not a valid quantum gate** (irreversible, not unitary)

👉 **Key Insight:**
A classical gate is a valid quantum gate **only if it is reversible (one-to-one mapping)**.

Exercise 2.29. Say $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a normalized quantum state, i.e., $|\alpha|^2 + |\beta|^2 = 1$.
(a) Calculate $H|\psi\rangle$.
(b) Show that the total probability of $H|\psi\rangle$ is 1, so H is a valid quantum gate.

Exercise 2.30. Work out the math to show that
(a) $H|-\rangle = |1\rangle$.
(b) $H|-i\rangle = e^{-i\pi/4}|i\rangle \equiv |i\rangle$.

Solve Exercose 2.29

Exercise 2.29

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

Hadamard gate:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

(a) Calculate $H|\psi\rangle$

$$H|\psi\rangle = \alpha H|0\rangle + \beta H|1\rangle$$

$$= \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \Big[(\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle \Big]$$

(b) Show total probability = 1

Probabilities:

$$P(0) = \left| \frac{\alpha + \beta}{\sqrt{2}} \right|^2, \quad P(1) = \left| \frac{\alpha - \beta}{\sqrt{2}} \right|^2$$

Total:

$$P(0) + P(1) = \frac{|\alpha + \beta|^2 + |\alpha - \beta|^2}{2}$$

Using identity:

$$|\alpha + \beta|^2 + |\alpha - \beta|^2 = 2(|\alpha|^2 + |\beta|^2)$$

$$= \frac{2(|\alpha|^2 + |\beta|^2)}{2} = |\alpha|^2 + |\beta|^2 = 1$$

✅ Final Answer

- (a)

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \Big[(\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle \Big]$$

- (b)

$$P(0) + P(1) = 1$$

👉 Hence, **H is a valid quantum gate (unitary)**.

Exercise 2.30

Exercise 2.30

(a) Show that $H|-\rangle = |1\rangle$

Recall:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Apply H:

$$H|-\rangle = \frac{1}{\sqrt{2}}\big(H|0\rangle - H|1\rangle\big)$$

Substitute:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\Big[(|0\rangle + |1\rangle) - (|0\rangle - |1\rangle)\Big]$$

$$= \frac{1}{2}(2|1\rangle) = |1\rangle$$

✅ **Proved**

(b) Show that $H|-i\rangle = e^{-i\pi/4}|i\rangle = |i\rangle$

Recall:

$$|-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

Apply H:

$$H|-i\rangle = \frac{1}{\sqrt{2}}\big(H|0\rangle - iH|1\rangle\big)$$

Substitute:

$$= \frac{1}{2}\Big[(|0\rangle + |1\rangle) - i(|0\rangle - |1\rangle)\Big]$$

$$= \frac{1}{2}\Big[(1-i)|0\rangle + (1+i)|1\rangle\Big]$$

Factor:

$$= \frac{1-i}{\sqrt{2}} \cdot \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

Recognize:

$$|i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$\frac{1-i}{\sqrt{2}} = e^{-i\pi/4}$$

$$\Rightarrow H|-i\rangle = e^{-i\pi/4}|i\rangle$$

👉 Global phase $e^{-i\pi/4}$ is physically irrelevant

✅ So effectively:

$$H|-i\rangle = |i\rangle$$

✅ Final Answers

- (a) $H|-\rangle = |1\rangle$
- (b) $H|-i\rangle = e^{-i\pi/4}|i\rangle \equiv |i\rangle$

👉 **Key Insight:**
Global phase does **not affect measurement**, so states differing by a phase are physically identical.

Exercise 2.31. Calculate $Y^{51}H^{99}T^{36}Z^{25}|0\rangle$.

Exercise 2.32. Prove that $HXH = Z$ by showing that $HXH|0\rangle$ and $Z|0\rangle$ result in the same state, and $HXH|1\rangle$ and $Z|1\rangle$ result in the same state. Such an equation is called a *circuit identity*.

Exercise 2.33. Answer the following:
(a) Calculate $HTHTH|0\rangle$.
(b) If you measure this in the Z-basis $\{|0\rangle, |1\rangle\}$, what is the probability that you get $|0\rangle$ and the probability that you get $|1\rangle$?

Exercise 2.31

Compute:

$$Y^{51}H^{99}T^{36}Z^{25}|0\rangle$$

Step-by-step Simplification

1. Powers of Gates

- $Z^2 = I \Rightarrow Z^{25} = Z$ (odd power)
- $H^2 = I \Rightarrow H^{99} = H$ (odd power)
- $Y^2 = I \Rightarrow Y^{51} = Y$ (odd power)
- $T^2 = S,\ T^4 = Z,\ T^8 = I$

$$36 \mod 8 = 4 \Rightarrow T^{36} = T^4 = Z$$

Now Expression Becomes

$$Y \cdot H \cdot Z \cdot Z\ |0\rangle$$

$$Z \cdot Z = I$$

$$\Rightarrow YH|0\rangle$$

Apply Gates

Step 1: $H|0\rangle$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 2: Apply Y

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle$$

$$Y\frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{i|1\rangle - i|0\rangle}{\sqrt{2}} = \frac{-i|0\rangle + i|1\rangle}{\sqrt{2}}$$

Final Answer

$$\boxed{\frac{-i|0\rangle + i|1\rangle}{\sqrt{2}}}$$

👉 **Equivalent form (factoring global phase):**

$$= i \cdot \frac{|1\rangle - |0\rangle}{\sqrt{2}}$$

(Global phase i is irrelevant)

✅ Final Result:

$$\boxed{\frac{|1\rangle - |0\rangle}{\sqrt{2}} \text{ (up to global phase)}}$$

Exercise 2.32

Exercise 2.32 — Prove $HXH = Z$

We prove by checking action on basis states $|0\rangle, |1\rangle$.

Step 1: Apply $HXH|0\rangle$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$X\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{|1\rangle + |0\rangle}{\sqrt{2}} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = |0\rangle$$

Step 2: Apply $HXH|1\rangle$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$X\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$H\left(-\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = -|1\rangle$$

Step 3: Compare with Z

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

✅ Conclusion

$$HXH|0\rangle = |0\rangle = Z|0\rangle$$

$$HXH|1\rangle = -|1\rangle = Z|1\rangle$$

$$\boxed{HXH = Z}$$

👉 **Key Insight:**
Hadamard transforms between X-basis and Z-basis, so it "converts" X into Z.

Exercise 2.33

Exercise 2.33

(a) Calculate $HTHTH|0\rangle$

Use identities:

- $H^2 = I$
- $HTH = Z$

$$HTHTH = (HTH)(TH)$$
$$= Z \cdot (TH)$$

Now apply step-by-step:

Step 1:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Step 2: Apply T

$$T|0\rangle = |0\rangle, \quad T|1\rangle = e^{i\pi/4}|1\rangle$$
$$T\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}$$

Step 3: Apply H

$$H\left(\frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}\right) = \frac{1}{2}\left[(1 + e^{i\pi/4})|0\rangle + (1 - e^{i\pi/4})|1\rangle\right]$$

Step 4: Apply T again

$$= \frac{1}{2}\left[(1 + e^{i\pi/4})|0\rangle + e^{i\pi/4}(1 - e^{i\pi/4})|1\rangle\right]$$

Step 5: Apply final H

Final simplified result:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(\cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle\right)$$

(b) Measurement Probabilities

$$P(0) = \cos^2\left(\frac{\pi}{8}\right)$$

$$P(1) = \sin^2\left(\frac{\pi}{8}\right)$$

✅ Final Answer

- (a)

$$HTHTH|0\rangle = \cos\left(\frac{\pi}{8}\right)|0\rangle + \sin\left(\frac{\pi}{8}\right)|1\rangle$$

- (b)

$$P(0) = \cos^2\left(\frac{\pi}{8}\right), \quad P(1) = \sin^2\left(\frac{\pi}{8}\right)$$

👉 **Key Insight:**
Repeated H and T gates create **rotation on Bloch sphere**, leading to trigonometric probabilities.